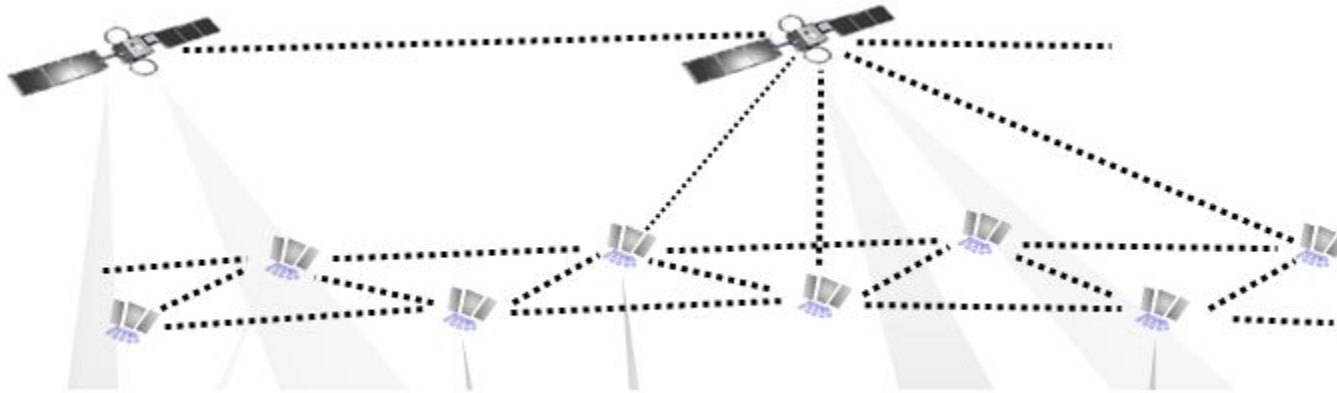


What is Satellite?



- (Satellite is a specialized antenna in a stable orbit above the earth.)
- In a satellite communications system, two or more stations on or near the earth communicate via one or more satellites that serve as relay stations in space.

Applications in Telecom

- global telephone connections
 - backbone for global networks
- replaced by fiber optics
- Reason:
 - Fiber optic link has tremendous capacity
 - Satellite link: signal need to travel 72000 km
 - Optical link: about 10,000 km while crossing Pacific or atlantic
 - Delay of 0.25 sec for Geostationary satellites
 - Connections for communication in remote places or underdeveloped areas
 - Global mobile communication
 - For data service by LEO
 - Tries to extend cellular network coverage, not to replace



Traditional vs. Modern Satellites

- Traditional satellite were simple transponders

- Receive a signal on one frequency
- Amplify it
- Transmit it on another frequency

a transponder is a device that, upon receiving a signal, emits a different signal in response. The term is a blend of transmitter and responder.

- Today: Satellites are like Flying routers

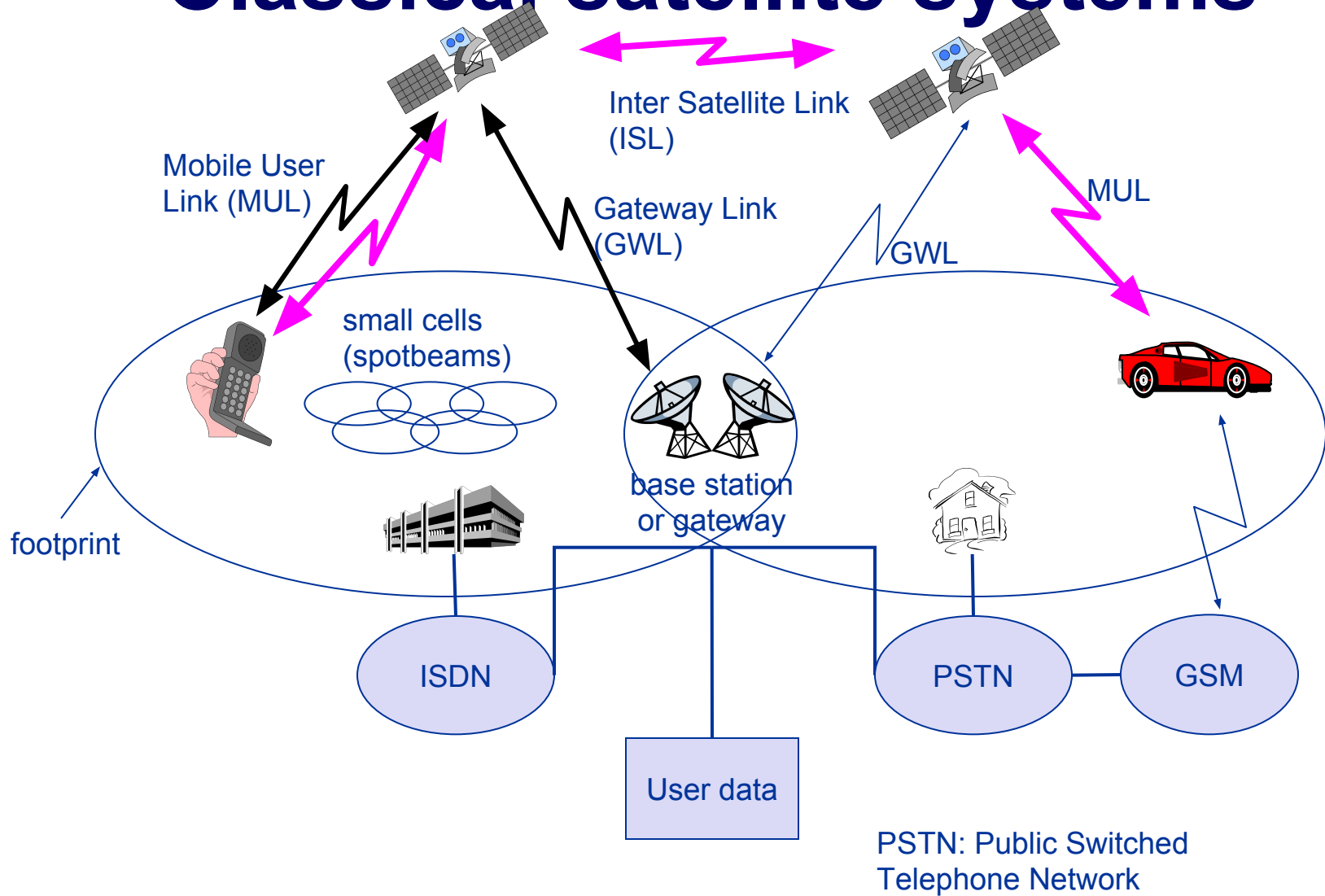
- Many functions of higher communication layers
 - Inter-satellite routing
 - Error correction

- Earlier only analog amplification

- Now digital regeneration also

- Satellite decodes the signals into a bit-stream
- And codes it again into a signal
 - Hence, higher quality of received signal on earth

Classical satellite systems



Satellite communication

Advantages:

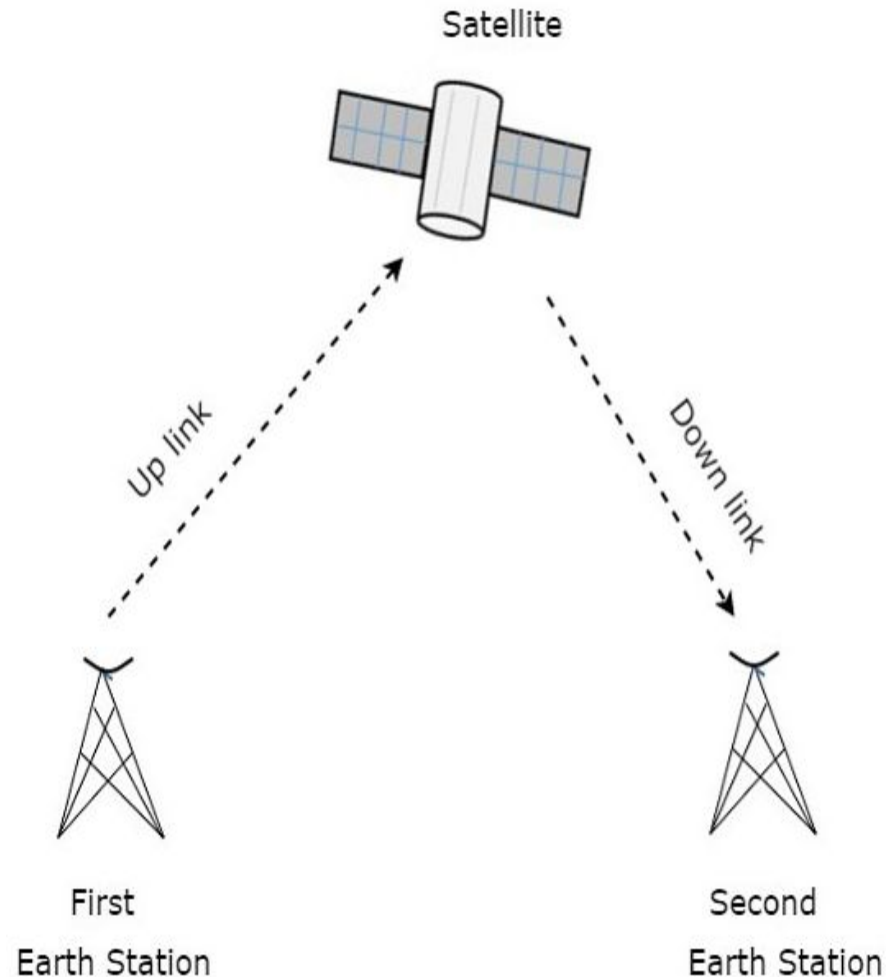
- Area of coverage is more than that of terrestrial systems. ✓
- Each and every corner of the earth can be covered. ✓
- Transmission cost is independent of coverage area. ✓
- Higher Bandwidths are available for use.

Disadvantages:

- Launching of satellites into orbits is a costly process. ✓
- Propagation delay of satellite systems is more than that of conventional terrestrial systems. ✓
- Difficult to provide repairing activities if any problem occurs in a satellite system. ✓
- Free space loss is more. ✓

Satellite-Related Terms

- The antenna systems on or near the earth are referred to as **earth stations**.
- A transmission from an earth station to the satellite is referred to as **uplink**, whereas transmissions from the satellite to the earth station are **downlink**.
- The component in the satellite that takes an uplink signal and converts it to a downlink signal is called a **transponder**.



Satellite-Related Terms

- **Inclination angle** is defined as the angle between the equatorial plane and the plane described by the satellite orbit.
 - An inclination angle of 0 degrees means that the satellite is exactly above the equator.
- **Elevation angle** is defined as the angle between the center of the satellite beam and the plane tangential to the earth's surface.
- ✓ **Footprint** can be defined as the area on earth where the signals of the satellite can be received.

Satellite Periods

The time required for a satellite to make a complete trip around the Earth, is determined by Kepler's law,

- which defines the period as a function of the distance of the satellite from the center of the Earth.



$$\text{Period} = C \times \text{distance}^{1.5}$$

Here C is a constant approximately equal to $1/100$. The period is in seconds and the distance in kilometers.



Example-1

What is the period of the Moon, according to Kepler's law?

$$e = e a^{3/2}$$

Solution

The Moon is located approximately 384,000 km above the Earth. The radius of the Earth is 6378 km. Applying the formula, we get.

$$\text{Period} = \frac{1}{100} (384,000 + 6378)^{1.5} = 2,439,090 \text{ s} = 1 \text{ month}$$



Example-2

According to Kepler's law, what is the period of a satellite that is located at an orbit approximately 35,786 km above the Earth?

Solution

e

Applying the formula, we get

$$\text{Period} = \frac{1}{100}(35,786 + 6378)^{1.5} = 86,579 \text{ s} = 24 \text{ h}$$

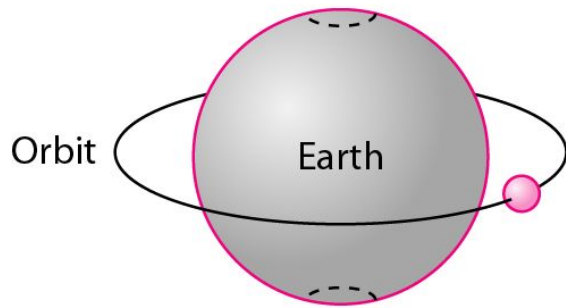


Example-2 (continued)

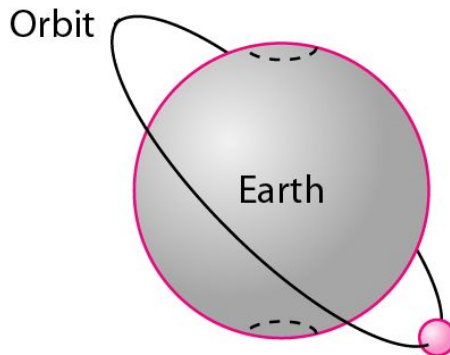
This means that a satellite located at 35,786 km has a period of 24 h, which is the same as the rotation period of the Earth. A satellite like this is said to be stationary to the Earth. The orbit, as we will see, is called a geosynchronous orbit.

Satellite orbits

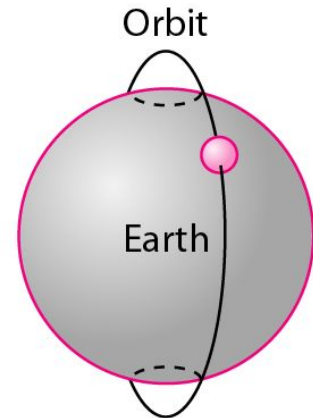
- An artificial satellite needs to have an orbit, the path in which it travels around the Earth.
- The orbit can be ✓
 - Equatorial- orbit above earth's equator
 - Polar- orbit passes over both poles
 - Inclined- Other orbits referred to as inclined orbits



a. Equatorial-orbit satellite



b. Inclined-orbit satellite

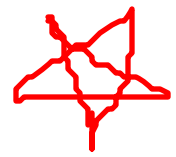


c. Polar-orbit satellite



Basics

- Satellites in circular orbits
 - attractive force of the earth due to gravity
$$F_g = m g (R/r)^2$$
 - Centrifugal force, $F_c = m r \omega^2$
 - m: mass of the satellite
 - R: radius of the earth ($R = 6370 \text{ km}$)
 - r: distance of the satellite to the center of earth
 - g: acceleration of gravity ($g = 9.81 \text{ m/s}^2$)
 - ω : angular velocity, $\omega = 2 \pi f$
 - f: rotation frequency



Basics

- To keep satellite stable in circular orbit

$$- F_g = F_c$$

$$- F_g = m g (R/r)^2$$

$$- F_c = m r \omega^2$$

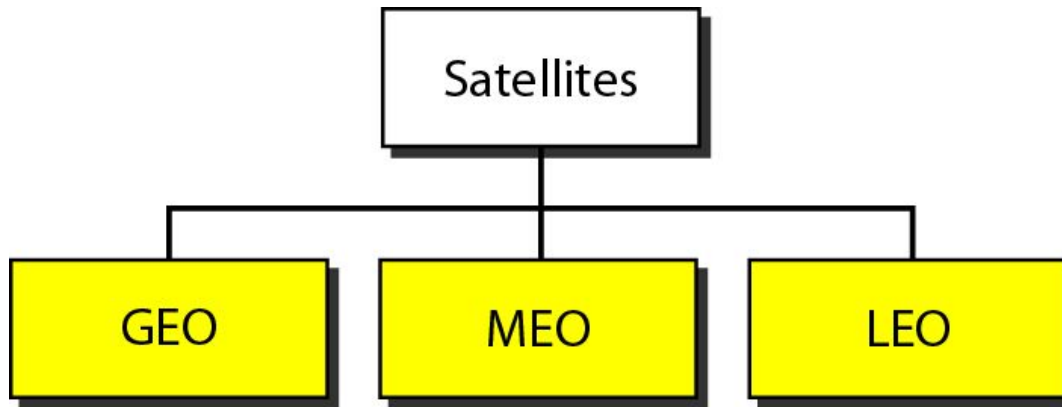
$$m g (R/r)^2 = m r \omega^2$$

- The distance r of the satellite to the center of the earth results in the following equation:

$$r = \sqrt[3]{\frac{gR^2}{(2\pi f)^2}}$$

Satellite categories

✓ Based on the location of the orbit, satellites can be divided into three categories: geostationary Earth orbit (GEO), low-Earth-orbit (LEO), and middle-Earth-orbit (MEO)



Satellite orbit altitudes

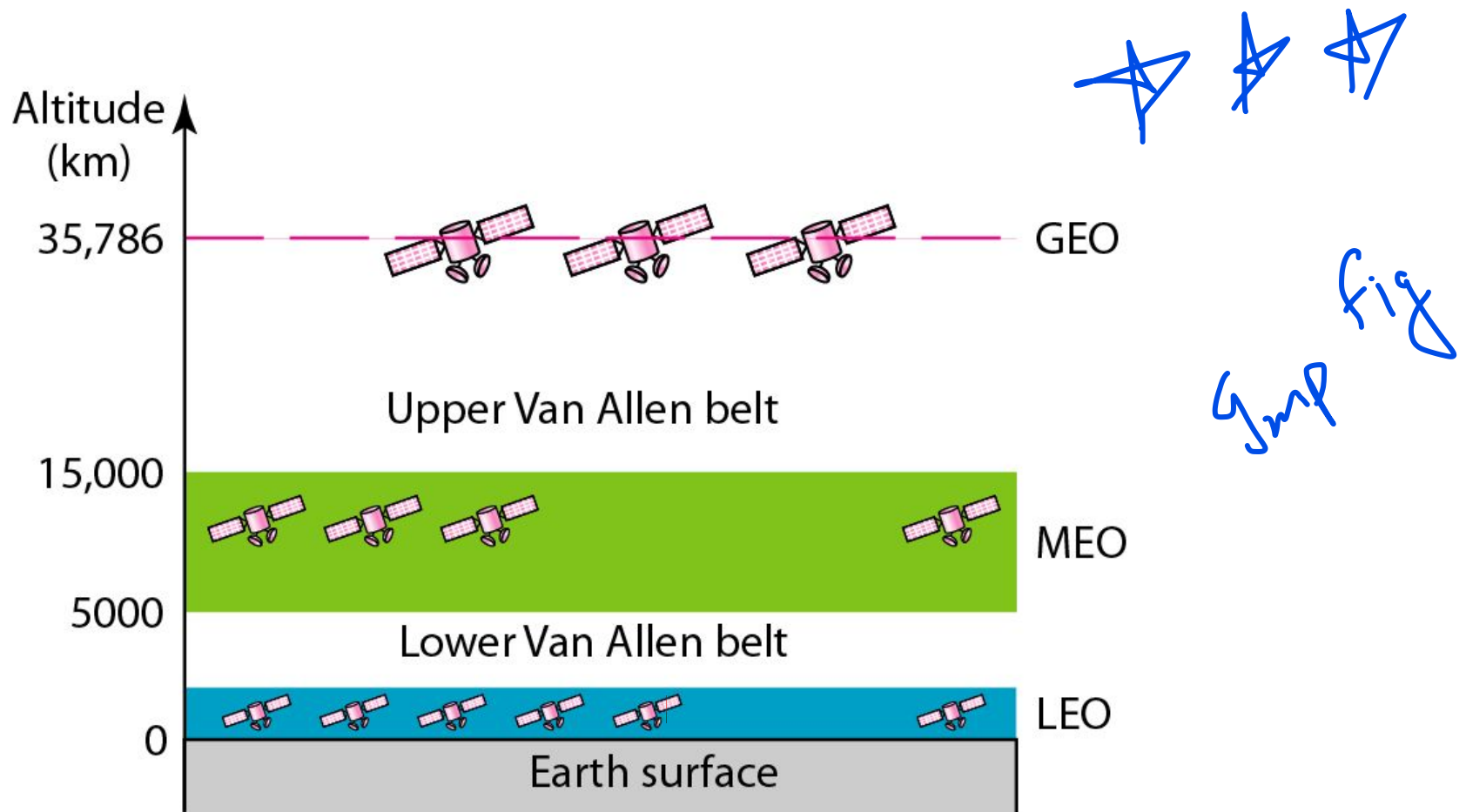


Figure shows the satellite altitudes with respect to the surface of the Earth. There is only one orbit, at an altitude of 35,786 km for the GEO satellite. MEO satellites are located at altitudes between 5000 and 15,000 km. LEO satellites are normally below an altitude of 2000 km. One reason for having different orbits is due to the existence of two Van Allen belts. A Van Allen belt is a layer that contains charged particles.



Geostationary satellites

- Orbit **35,786 km** distance from earth surface, orbit in equatorial plane (inclination 0°)
- complete rotation **exactly one day**, satellite is synchronous to earth rotation
- fix antenna positions, no adjusting necessary
- A large footprint (up to 34% of earth surface!), therefore difficult to reuse frequencies
- Three satellites can cover the whole globe
- HIGH transmit power needed
- high latency due to long distance (275 ms)
- Not useful for global coverage **for small mobile phones** and data transmission, **typically used for radio and TV transmission**



LEO systems

- Orbit 500 - 1500 km above earth surface
- visibility of a satellite 95 - 120 minutes
- global radio coverage possible
- latency comparable with terrestrial long distance connections, 5 - 10 ms
- smaller footprints, **better frequency reuse**
- but now **HANDOVER necessary** from one satellite to another
- many satellites necessary for global coverage
- more complex systems due to moving satellites
- Examples:
 - Iridium (start 1998, 66 satellites)
 - Globalstar (start 1999, 48 satellites)





MEO systems

- Orbit approx. 5000 - 12000 km above earth surface
- Comparison with LEO systems:
 - slower moving satellites
 - less satellites needed
 - simpler system design
 - for many connections no hand-over needed
 - higher latency, about 70 - 80 ms
 - higher sending power needed
 - special antennas for small footprints needed
- Example:
 - GPS: period 12 hrs

Table-1: *Satellite frequency bands*

<i>Band</i>	<i>Downlink, GHz</i>	<i>Uplink, GHz</i>	<i>Bandwidth, MHz</i>
L	1.5	1.6	15
S	1.9	2.2	70
C	4.0	6.0	500
Ku	11.0	14.0	500
Ka	20.0	30.0	3500